THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · MEDIUM LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. What is the sum of the first 12 even numbers $2 + 4 + 6 + \ldots$? 1

(a) 144 **(b)** 156 **(c)** 132 **(d)** 78

2. The hour hand of a clock turns through 150°. How many hours have passed? 1

(a) 2 **(b)** 7 **(c)** 5 **(d)** 10

3. The greatest 5-digit palindrome with first digit 2 is: 1

(a) 99999 **(b)** 29892 **(c)** 2992 **(d)** 29992

4. A bar of height 6 cm in a graph with scale 1 cm = 15 units, and another bar of 5 cm. Together they show: 1

(a) 165 **(b)** 150 **(c)** 26 **(d)** 180

5. The LCM of 12 and 18 is: 1

(a) 18 **(b)** 36 **(c)** 72 **(d)** 216

6. A rectangular room 8 m by 5 m has a carpet costing ₹120 per m^2. The cost of carpeting (₹): 1

(a) 4900 **(b)** 4680 **(c)** 4800 **(d)** 4920

7. $\frac{3}{5}$ of a number is 36. The number is: 1

(a) 60 **(b)** 63 **(c)** 72 **(d)** 48

8. A rectangle's diagonals meet at O making $\angle AOB = 50^\circ$. Then $\angle AOD$ is (degrees): 1

(a) 130 **(b)** 160 **(c)** 120 **(d)** 140

9. A figure has 4 lines of symmetry. Which could it be? 1

(a) A scalene triangle **(b)** A rhombus (not a square)
(c) A square **(d)** A rectangle

10. The sum of all integers from -4 to 4 is: 1

(a) -1 **(b)** 0 **(c)** 1 **(d)** 4

11. Which triangular number comes just after 190? 1

(a) 231 **(b)** 209 **(c)** 210 **(d)** 211

12. A reflex angle is 200°. The angle needed to complete a full turn is: 1

(a) 110 **(b)** 160 **(c)** 20 **(d)** 180

13. Starting from 16, the hailstone (Collatz) sequence reaches 1 in how many steps? 1

(a) 4 **(b)** 3 **(c)** 5 **(d)** 6

14. What fraction of the total do Hockey players form? 1

Sport	Students
Hockey	30
Kabaddi	35
Tennis	65
Chess	15

(a) $\frac{1}{5}$ **(b)** $\frac{6}{23}$ **(c)** $\frac{6}{29}$ **(d)** $\frac{23}{29}$

15. The smallest digit to put in the blank to make 452_ divisible by 3 is: 1

(a) 2 **(b)** 1 **(c)** 3 **(d)** 5

16. An equilateral triangle of side 12 cm and a square have the same perimeter. The side of the square (cm) is: 1

(a) 15 **(b)** 9 **(c)** 12 **(d)** 6

17. After spending $\frac{3}{8}$ of her money, Neha has ₹100 left. What fraction is left? 1

(a) $\frac{3}{8}$ **(b)** $2\frac{2}{3}$ **(c)** $\frac{5}{9}$ **(d)** $\frac{5}{8}$

18. A circle has diameter 24 cm. A concentric circle has radius 7 cm. The width of the ring (cm) is: 1

(a) 7 **(b)** 4 **(c)** 6 **(d)** 5

19. A square is rotated about its centre through $30°$. The smallest further turn (degrees) after which it again coincides with itself is: 1

(a) 70 **(b)** 150 **(c)** 60 **(d)** 50

20. The difference between the highest and lowest of -8, 5, -3, 9 is: 1

(a) 17 **(b)** 19 **(c)** 16 **(d)** 18

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Find $5\frac{3}{4} - 2\frac{1}{2}$. 2

22. Two concentric circles have radii 4 cm and 9 cm. Find the width of the ring between them, and the diameters of both circles. 2

23. A regular polygon rotates through $40°$ to coincide with itself. How many sides does it have? Name its order of symmetry. 2

24. Find $-18 + 25 + (-9)$. 2

25. Is 36 a term of $3, 7, 11, 15, \ldots$? Give a reason. 2

26. OB bisects $\angle AOC$ and $\angle AOB = 28°$. Find $\angle AOC$ and its type. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. A regular polygon has 5 lines of symmetry. Find its angle of rotation, the angle after 3 turns and the number of turns to make a full rotation. 3

28. A train station has platforms numbered with 0 at ground level. A lift moves 3 floors below ground, then 5 floors up, then 4 floors below again. Find its final floor and the total number of floors moved. 3

29. Find the sum of all odd numbers from 1 to 59. Hence find the sum of all even numbers from 2 to 60. 3

30. A ray OA is turned about O: first 60° anticlockwise, then 70° anticlockwise, then 40° clockwise. Find the net angle of turn from the original position. 3

31. Find the largest palindrome less than 1300 and the smallest palindrome greater than 1300. 3

OR

A shop sold 40, 60, 80 and 100 kg of rice in four weeks. Find the total. For a bar graph with the scale 1 cm = 20 kg, find the height of each bar. 3

32. Two bells ring every 12 minutes and every 18 minutes. They ring together at 9 am. When do they next ring together? 3

OR

Two rectangles have the same perimeter 48 cm: one is 20 cm by 4 cm and the other is 12 cm by 12 cm. Which has the larger area and by how much? 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. A frog is on the number line at -3. It jumps 8 units to the right, then 5 units to the left, then 9 units to the left. Find its final position, and the distance from 0. 5

34. Show that the sum of the cubes of the first 9 numbers is a perfect square, and find the sum of the first 9 numbers. 5

35. Four rays OA, OB, OC, OD are drawn from O so that $\angle AOB = 30^\circ$, $\angle BOC = 45^\circ$ and $\angle COD = 60^\circ$, measured one after the other in the same direction. Find $\angle AOD$ and the angle left to complete a straight angle. 5

36. Using the digits 3, 7, 5, 1 once each, form a 4-digit number closest to 5000. Show the options you tried and the difference. 5

OR

A school survey data is given. Find the total, the sport with maximum students, and the fraction who play it. Then choose a scale for a bar graph and give the height of the biggest bar. Data: Kabaddi 60, Hockey 20, Tennis 35. 5

37. Find the smallest number that leaves a remainder 1 when divided by 4, 5 and 6, and the next such number. 5

OR

A square of side 20 cm has a square of side 3 cm cut from each of its four corners. Find the area of the remaining figure and its perimeter. 5

38. A tank is $\frac{1}{4}$ full. After 20 litres are added, it becomes $\frac{3}{4}$ full. Find the capacity of the tank. 5

OR

A gardener draws two concentric circles of radii 5 m and 8 m. Find the width of the path between them, and the diameters of the outer and inner circles. 5

END OF QUESTION PAPER — Total: 80 Marks

MASTERMIND